Algebraic and Differential Topology – Lecture Notes

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1 Introduction

Definition 1.1. A topology on a set X is a collection τ of subsets of X such that:

(i) $\varnothing, X \in \tau$.

(ii) If $U_i \in \tau$ for all $i \in I$, then $\bigcup_{i \in I} U_i \in \tau$.

(iii) If $U_i \in \tau$ for all $i \in I$ where I is finite, then $\bigcap_{i \in I} U_i \in \tau$.

We call the pair (X, τ) a topological space. The elements of τ are called open sets.

Definition 1.2. A **basis** for a topology on X is a collection $\mathcal{B}$ of subsets of X such that:

(i) $\forall x \in X$, $\exists B \in \mathcal{B}$ containing x.

(ii) If $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \mathcal{B}$, then there is a basis element $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

$\mathcal{B}$ generates a topology τ on X as follows:
$U \in \tau$ if $\forall x \in U$, $\exists B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Remark 1.3. It is important that $B \in \mathcal{B}$ is a subset of U in the last statement. If we only required $x \in B \cap U$, we would run into problems. For example consider $X = \mathbb{R}$ and $U = \{0, 1\}$. We know that $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}, a < b\}$ is a basis for the standard topology on $\mathbb{R}$. Then for $x = 0$, we can choose $B = (-1, 1)$ and for $x = 1$, we can choose $B = (0, 2)$. This would mean that U is open, which should not be the case.

Example 1.4.

- (M, d) is a metric space. Then $\mathcal{B} = \{B_r(x) : x \in M, r > 0\}$ is a basis for the metric topology on M, where $B_r(x) = \{y \in M : d(x, y) < r\}$ is the open ball of radius r around x.
- X, Y topological spaces. The product topology on $X \times Y$ is generated by the basis $\mathcal{B} = \{U \times V : U \subseteq X, V \subseteq Y \text{ open}\}$.
- subspace topology: Let (X, τ) be a topological space, $S \subseteq X$. Then the subspace topology of S is $\tau_S = \{U \cap S : U \in \tau\}$.

Definition 1.5. U is a neighbourhood of x if $\exists V$ open such that $x \in V \subseteq U$.

Definition 1.6. Let X, Y be topological spaces.
A map $f : X \to Y$ is continuous if $\forall U \subseteq Y$ open, $f^{-1}(U)$ is open in X.
If f is a bijection and both f and f^{-1} are continuous, then f is called a homeomorphism. In this case, we say that X and Y are homeomorphic or topologically equivalent ($X \simeq Y$).

Example 1.7.

- $f : [0,1] \to \mathbb{S}^1$ given by $f(t) = (cos(2\pi t), sin(2\pi t))$.
  f is continuous and bijective but f^{-1} is not continuous.
  $f([0,1))$ is not open in $\mathbb{S}^1$.

- $\mathbb{R} \not\cong \mathbb{R}^2$: Assume they are homeomorphic. Then there exists a homeomorphism $f : \mathbb{R} \to \mathbb{R}^2$ with $f(0) = (a,b)$. But $f|_{\mathbb{R}\setminus\{0\}} : \mathbb{R} \setminus \{0\} \to \mathbb{R}^2 \setminus \{(a,b)\}$ is a homeomorphism. But $\mathbb{R} \setminus \{0\}$ is not connected, while $\mathbb{R}^2 \setminus \{(a,b)\}$ is connected. This is a contradiction.

- $\mathbb{R}^2 \not\cong \mathbb{R}^3$: Assume they are diffeomorphic. Then there exists a diffeomorphism $f : \mathbb{R}^2 \to \mathbb{R}^3$. Then $f \circ f^{-1} = id_{\mathbb{R}^3}, f^{-1} \circ f = id_{\mathbb{R}^2}$ so $Df_x Df_y^{-1} = I_3$ and $Df_y^{-1} Df_x = I_2$.

2 Homotopy

Definition 2.1. Let X, Y be topological spaces and $f, g : X \to Y$ be continuous maps. Then f is called homotopic to g if there exists a $F : X \times I \to Y$ continuous s.t. $F(x,0) = f(x)$ and $F(x,1) = g(x)$ for all $x \in X$. We write $f \simeq_F g$ and F is called a homotopy between f and g. If $f \simeq c$ where c is a constant map, then f is called null-homotopic.

Definition 2.2. Let X be a topological space. A continuous map $f : I \to X$ is called a path in X.
Two paths $f, g : I \to X$ are called **path homotopic** if they have the same start and end points, i.e. $f(0) = g(0)$ and $f(1) = g(1)$, and there exists a continuous map $F : I \times I \to X$ s.t. $F(x,0) = f(x)$, $F(x,1) = g(x)$ for all $x \in I$, and $F(0,t) = f(0) = g(0)$, $F(1,t) = f(1) = g(1)$ for all $t \in I$. We write $f \simeq_F g$.

Remark 2.3. This is an equivalence relation:

(i) $f \simeq_F f$ with $F(x,t) = f(x)$ for all $x, t \in I$.

(ii) $f \simeq_F g \implies g \simeq_G f$ with $G(x,t) := F(x, 1-t)$.

(iii) $f \simeq_F g$, $g \simeq_G h \implies f \simeq_H h$ with $\begin{cases} H(x,t) = F(x,2t) & 0 \le t \le \frac{1}{2} \\ H(x,t) = G(x,2t-1) & \frac{1}{2} \le t \le 1 \end{cases}$

If f is a path then we call $[f]$ the path homotopy equivalence class of f.